

Software Test Plan (STP) - Leave Management System (LMS)

Project: Automated Leave Management System (LMS)

Version: 1.0

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Date: 13 -09-2025

Status: Version 1

1. Introduction

1.1 Purpose:

This document defines the test plan for the Automated Leave Management System v1.0. It outlines the objectives, scope, strategy, resources, schedule, and responsibilities for testing to ensure the system functions as intended. The purpose of testing is to validate that the system accurately manages employee leave requests, approvals, balances, and reporting, while ensuring reliability, usability, and compliance with organizational policies.

1.2 Scope:

Testing covers key features of the Automated Leave Management System, including user authentication, leave application submission, approval workflows, leave balance calculation, notifications, and reporting. The focus is on validating functional correctness, usability, and compliance with organizational leave policies. Integration with external HR or payroll systems, as well as underlying infrastructure and hardware maintenance, are excluded from the scope.

1.3 References:

Automated Leave Management System SRS v1.0, Design Specifications v1.0, Organizational HR Policies and Leave Guidelines, Software Quality Standards (e.g., ISO/IEC 25010), Database Design Document v1.0

1.4 Definitions:

LMS (Leave Management System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), HR (Human Resources), Leave Balance, Workflow.

2. Test Items

- User Authentication Module
- Leave Application Module
- Leave Approval Workflow Module
- Leave Balance Management Module
- Notification and Alerts Module
- Reporting and Analytics Module
- HR Management Interface

3. Features to be Tested

Features mapped to SRS requirement IDs:

- **LMS-F-001:** Secure user login and role assignment with encryption.
- **LMS-F-011:** Leave request submission and validation (dates, type, balance)
- **LMS-F-014:** Leave approval/rejection workflow (Manager/HR)
- **LMS-F-021:** Leave balance maintenance and updates
- **LMS-F-031:** HR policy setup (leave types, accrual rules, carry-forward, holiday calendar)
- **LMS-F-041:** Leave cancellation workflow (request, approval/rejection)
- **LMS-F-071:** Notification and alert generation (employee, manager, real-time delivery)
- **LMS-F-081:** Reporting and analytics (leave history, balance usage, filters, export)
- **LMS-F-056:** Audit logging of login/logout activities
- **PRJ-SR-001:** Weekly automated backups and restore capability
- **PRJ-SR-004:** Role-Based Access Control (RBAC)
- **PRJ-SR-002:** Secure leave record protection and authorization controls.

4. Features Not to be Tested

- Integration with third-party HRMS/Payroll or ERP systems (assumed tested separately)
- Underlying database engine performance and optimization (vendor responsibility)
- Operating system and server infrastructure reliability

- External email/SMS gateway services (only system trigger will be tested, not the third-party delivery)
- Hardware or network infrastructure dependencies

5. Test Approach / Strategy

Levels:

- **Unit Tests:** Verification of individual modules (e.g., authentication, leave application, approval workflow).
- **Integration Tests:** Validation of interactions between modules (e.g., leave application ↔ balance calculation ↔ approval).
- **System Tests:** End-to-end testing of complete LMS functionality in a controlled environment.
- **Acceptance Tests (UAT):** Validation by HR and selected employees to ensure business requirements are met.

Types:

- **Functional Testing:** Verification of core features such as leave requests, approvals, balance tracking, notifications, and reporting.
- **Regression Testing:** Ensuring existing functionality remains stable after changes or updates.
- **Performance Testing:** Measuring response time under typical and peak loads, ensuring scalability.
- **Usability Testing:** Assessing ease of use, clarity of the user interface, and accessibility for end users.
- **Security Testing:** Validating role-based access, data confidentiality, and protection against unauthorized access.

Entry Criteria:

- Stable build of LMS delivered for testing.
- Test environment set up and configured.
- Test data (employee records, leave policies) prepared and available.
- All required documents (SRS, design specs) accessible.

Exit Criteria:

- 100% of planned test cases executed successfully.
- No open critical or high-severity defects.
- All functional and non-functional acceptance criteria satisfied.
- Test summary report reviewed and approved by stakeholders.

5.1 Security Validation

- Validate secure login and password handling (masked input, no plain-text storage/logging).
- Verify role-based access control for Employee, Manager, HR, and Admin roles.
- Ensure data encryption in transit (TLS 1.2+) and at rest.
- Verify audit logging of authentication events (login, logout, failed attempts).
- Ensure compliance with organizational IT security and data protection policies (e.g., GDPR, ISO 27001).

6. Test Environment

Hardware:

- Standard client machines (desktops/laptops) with browsers
- Application server and database server hosting the LMS

Software:

- Database (MySQL)
- Web server / Application server
- Web Browser

Tools:

- **Cypress** – for UI automation testing
- **Postman** – for API testing
- **Jira** – for defect tracking
- **SonarQube** – for code quality and security testing

Test Data:

- Dummy employee records (employee IDs, departments, roles)
- Sample leave policies (sick leave, earned leave, maternity leave)
- Test leave applications (valid, invalid, overlapping, excess leave)
- Predefined approval workflows (employee → manager → HR)

7. Test Schedule

Milestones:

- Test case design: 29-Sep-2025
- Environment setup: 05-Oct-2025
- Test execution start: 07-Oct-2025
- Test execution end: 16-Oct-2025
- UAT: 22-Oct-2025 to 27-Oct-2025

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Deepthi	Prepare plan, coordinate execution.
Test Engineer	Hita	Design & execute test cases, log defects
Test engineer	Chirag	Design & execute test cases, log defects
Developer	Samrudhi	Support defect fixes and triage
Product Owner	Team_LMS	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Data loss/corruption	Implement weekly automated backups and test the restore process.
Unauthorized access	Role-based authentication (Admin, Faculty, Student).
User adoption issues	Provide simple UI; training/demo sessions; clear user guide.

11. Assumptions & Dependencies

- HRMS API sandbox is available and stable
- Employee and leave type data is preloaded
- Notification gateway credentials available

12. Suspension & Resumption Criteria

Suspend testing if:

- Test the environment (LMS application, database, or API sandbox) is unavailable for more than 4 hours.
- The build is too unstable (blocking defects affect more than 30% of planned test cases).
- Critical services required for LMS functionality are down (e.g., notification gateway, HRMS API sandbox).

Resume testing if:

- All blocking defects preventing test execution are resolved.
- Test the environment and required external services (HRMS API sandbox, notification gateway) are stabilized and fully functional.

13. Test Case Management & Traceability

Requirements mapped to test cases via RTM:

- **LMS-F-001** → TC-Auth-01, TC-Auth-02
- **LMS-F-011** → TC-LM-01, TC-LM-02
- **LMS-F-014** → TC-Perf-01
- **LMS-F-031**-TC-Auth-03
- **LMS-F-056**-TC-LM-03
- **PRJ-SR-002**-TC-Sec-01
- **PRJ-SR-003** → TC-Sec-02, TC-Sec-03
- **PRJ-SR-004**-TC-Sec-04

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density

- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Deepthi M	<i>Deepthi</i> 13/09/25
Test engineer	Hita R	<i>Hita</i> 13/09/25
Test engineer	Chirag G D	Chirag 13/09/25
Dev Lead	Samrudhi Rao	<i>Samrudhi</i> 13/09/25
Product Owner	Team_LMS	13/09/25

